

Web Application Development

IS4102



Lecture Note 06 - IP Address

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What is an IP Address?

IP = Internet Protocol Address

A unique number given to a device on a network

Helps devices send and receive data correctly

Example: 192.168.1.1

Home address analogy

Just like a home address tells a delivery person where to go, an IP address tells data where to go on a network.



Purpose of an IP Address

Identification

Recognizes each device on a network.

Communication

Allows devices to exchange messages and files.

Routing

Helps routers choose the correct path for packets.

Internet Access

Needed for browsing websites and using online services.

Troubleshooting

Used to find network issues and device location in a network.



Types of IP Addresses

Public IP

Used on the internet.
Your router normally
has one.

Private IP

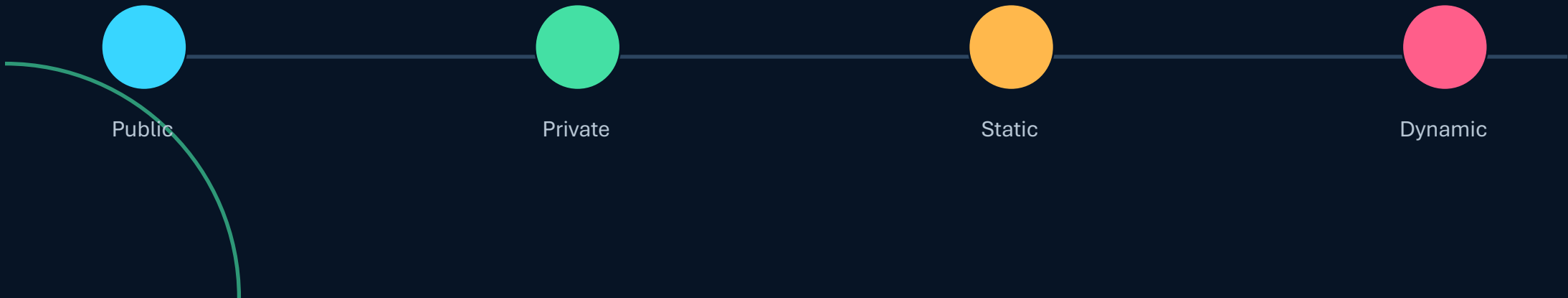
Used inside homes,
schools, offices and
local networks.

Static IP

Stays fixed. Common
for servers and
important devices.

Dynamic IP

Changes
automatically, usually
assigned by DHCP.



IPv4: The Common Format

IPv4 uses four decimal numbers separated by dots.



32-bit address system
Around 4.3 billion possible addresses
Still widely used today
Address shortage led to IPv6

IPv4 Example

192.168.0.1

Each number can range from 0 to 255.

Think of it as an older address system that still works, but has limited space.

IPv6: The Newer Format

IPv6 uses 128 bits, giving a much larger address space.

2001:0db8:85a3::8a2e:0370:7334

Written in hexadecimal

Designed to solve IPv4 address shortage

Useful for modern internet growth and many connected devices

128-bit

IPv4



IPv6



IPv6 gives the internet much more room to grow.

How IP Addresses Work in Web Applications



Enter URL

User types a website address into the browser.



DNS Lookup

DNS translates the domain name into an IP address.



Connect

Browser connects to the server at that IP address.



Response

Server sends the webpage data back to the browser.

Example:

`www.example.com` → DNS → `93.184.216.34` → Server → Webpage Loaded

Conclusion

Key Takeaways

An IP address is the digital identity of a device
It makes communication and routing possible
IPv4 is common, but limited
IPv6 supports the future growth of the internet

Without IP addresses, the internet would not know where to deliver data.

THANK YOU

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